J2 Vector magnitude and direction

Name:

Class:

Date:

Time: **402 minutes**

Marks: **334 marks**

Comments:

Q1.

A quadrilateral has vertices A, B, C and D with position vectors given by

$$\overrightarrow{OA} = \begin{bmatrix} 3 \\ 5 \\ 1 \end{bmatrix}, \overrightarrow{OB} = \begin{bmatrix} -1 \\ 2 \\ 7 \end{bmatrix}, \overrightarrow{OC} = \begin{bmatrix} 0 \\ 7 \\ 6 \end{bmatrix} \text{ and } \overrightarrow{OD} = \begin{bmatrix} 4 \\ 10 \\ 0 \end{bmatrix}$$

(a) Write down the vector $\overrightarrow{AB}$

(1)

(b) Show that *ABCD* is a parallelogram, but not a rhombus.

(5)

(Total 6 marks)

Q2.

(a) The unit vectors **i** and **j** are perpendicular.
Find the magnitude of the vector −20**i** + 21**j**
Circle your answer.

-1 $\quad$ 1 $\quad$ $\sqrt{41}$ $\quad$ 29

(1)

(b) The angle between the vector **i** and the vector −20**i** + 21**j** is θ
Which statement about θ is true?
Circle your answer.

$0° < \theta < 45°$ $\quad$ $45° < \theta < 90°$ $\quad$ $90° < \theta < 135°$ $\quad$ $135° < \theta < 180°$

(1)

(Total 2 marks)

Q3.

The three forces $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$ are acting on a particle.

$\mathbf{F}_1 = (25\mathbf{i} + 12\mathbf{j})$ N
$\mathbf{F}_2 = (-7\mathbf{i} + 5\mathbf{j})$ N
$\mathbf{F}_3 = (15\mathbf{i} - 28\mathbf{j})$ N

The unit vectors **i** and **j** are horizontal and vertical respectively.
The resultant of these three forces is **F** newtons.

(a) (i) Find the magnitude of **F**, giving your answer to three significant figures.

(2)

(ii) Find the acute angle that **F** makes with the horizontal, giving your answer to the nearest 0.1°

(2)

(b) The fourth force, $\mathbf{F}_4$, is applied to the particle so that the four forces are in equilibrium.
Find $\mathbf{F}_4$, giving your answer in terms of **i** and **j**.

(1)
(Total 5 marks)

Q4.

In this question use g = 9.81 m s^{-2}, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector (40**i** – 10**j**) metres.
The unit vectors **i** and **j** are horizontal and vertical respectively.
Assume that there are no resistance forces acting on the ball.

(a) Find the speed of the ball when it is at a height of 3 metres above its initial position.

(6)

(b) State the speed of the ball when it is at its maximum height.

(1)

(c) Explain why the answer you found in part **(b)** may not be the actual speed of the ball when it is at its maximum height.

(1)
(Total 8 marks)

Q5.

Three forces act on a particle. These forces are (9**i** – 3**j**) newtons, (5**i** 8**j**) newtons and (–7**i** + 3**j**) newtons. The vectors **i** and **j** are perpendicular unit vectors.

(a) Find the resultant of these forces.

(2)

(b) Find the magnitude of the resultant force.

(2)

(c) Given that the particle has mass 5 kg, find the magnitude of the acceleration of the particle.

(2)

(d) Find the angle between the resultant force and the unit vector **i**.

(3)
(Total 9 marks)

Q6.

A particle, of mass 10 kg, moves on a smooth horizontal plane. At time t seconds, the acceleration of the particle is given by

$$\{(40t + 3t^2)\mathbf{i} + 20e^{-4t}\,\mathbf{j}\} \text{ m s}^{-2}$$

where the vectors **i** and **j** are perpendicular unit vectors.

(a) At time $t = 1$, the velocity of the particle is $(6\mathbf{i} - 5e^{-4}\mathbf{j})$ m s^{-1}.

Find the velocity of the particle at time t.

(5)

(b) Calculate the initial speed of the particle.

(3)

(Total 8 marks)

Q7.

A particle, of mass 50 kg, moves on a smooth horizontal plane. A single horizontal force

$$[(300t - 60t^2)\mathbf{i} + 100e^{-2t}\,\mathbf{j}] \text{ newtons}$$

acts on the particle at time t seconds.

The vectors **i** and **j** are perpendicular unit vectors.

(a) Find the acceleration of the particle at time t.

(2)

(b) When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j})$ m s^{-1}.

Find the velocity of the particle at time t.

(4)

(c) Calculate the speed of the particle when $t = 1$.

(4)

(Total 10 marks)

Q8.

A particle moves with a constant acceleration of $(0{:}1\mathbf{i} - 0{:}2\mathbf{j})$ m s^{-2}. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j})$ m s^{-1}. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

(b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

(c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q9.

A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j})$ m s^{-2}. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j})$ m s^{-1}. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the position vector of the particle t seconds after it has left the origin.

(2)

(b) Find the time that it takes for the particle to reach the point where it is due east of the origin.

(3)

(c) Find the speed of the particle when it is travelling south-east.

(6)

(Total 11 marks)

Q10.

A particle moves with constant acceleration $(-0.4\mathbf{i} + 0.2\mathbf{j})$ m s^{-2} . Initially, it has velocity $(4\mathbf{i} + 0.5\mathbf{j})$ m s^{-1} . The unit vectors **i** and **j** are directed east and north respectively.

(a) Find an expression for the velocity of the particle at time t seconds.

(2)

(b) (i) Find the velocity of the particle when $t = 22.5$.

(2)

(ii) State the direction in which the particle is travelling at this time.

(1)

(c) Find the time when the speed of the particle is 5 m s^{-1}.

(6)

(Total 11 marks)

Q11.

The velocity of a particle at time t seconds is **v** m s^{-1} , where

$$\mathbf{v} = (4 + 3t^2)\mathbf{i} + (12 - 8t)\mathbf{j}$$

(a) When $t = 0$, the particle is at the point with position vector $(5\mathbf{i} - 7\mathbf{j})$ m.

Find the position vector, **r** metres, of the particle at time t.

(4)

(b) Find the acceleration of the particle at time t.

(2)

(c) The particle has mass 2 kg.

Find the magnitude of the force acting on the particle when $t = 1$.

(4)
(Total 10 marks)

Q12.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j})$ m s^{-2}. The helicopter does not change its height above sea level as it moves. The unit vectors **i** and **j** are directed east and north respectively.

(a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse.

(4)

(b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree.

(3)

(c) The helicopter stops accelerating when it is 500 metres from its initial position.

Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)
(Total 12 marks)

Q13.

A boat moves with constant acceleration, so that its velocity **v** m s^{-1} at time t seconds is given by

$$\mathbf{v} = (4.5 - 0.9t)\mathbf{i} + (2 + 0.5t)\mathbf{j}$$

where the unit vectors **i** and **j** are directed east and north respectively.

(a) Find the velocity of the boat when $t = 0$.

(1)

(b) Find the velocity of the boat when $t = 5$.

(1)

(c) State the direction in which the boat is travelling when $t = 5$.

(1)

(d) Find the acceleration of the boat.

(3)

(e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat.

(3)

(f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector **i**.

(4)

(Total 13 marks)

Q14.

The constant forces $\mathbf{F}_1 = (8\mathbf{i} + 12\mathbf{j})$ newtons and $\mathbf{F}_2 = (4\mathbf{i} - 4\mathbf{j})$ newtons act on a particle. No other forces act on the particle.

(a) Find the resultant force acting on the particle.

(2)

(b) Given that the mass of the particle is 4 kg, show that the acceleration of the particle is $(3\mathbf{i} + 2\mathbf{j})$ m s^{-2}.

(2)

(c) At time t seconds, the velocity of the particle is $\mathbf{v}$ m s^{-1}.

(i) When $t = 20$, $\mathbf{v} = 40\mathbf{i} + 32\mathbf{j}$.

Show that $\mathbf{v} = -20\mathbf{i} - 8\mathbf{j}$ when $t = 0$.

(3)

(ii) Write down an expression for $\mathbf{v}$ at time t.

(1)

(iii) Find the times when the speed of the particle is 8 m s^{-1}.

(6)

(Total 14 marks)

Q15.

A ship moves with constant acceleration. At time t seconds, its velocity is $\mathbf{v}$ m s^{-1}, where

$$\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

(a) Write down the velocity of the ship when $t = 0$.

(1)

(b) Find the acceleration of the ship.

(2)

(c) Find t when the ship is travelling south-east.

(3)

(Total 6 marks)

Q16.

A particle, of mass 10 kg, moves on a smooth horizontal surface. A single horizontal force, $(9\mathbf{i} + 12\mathbf{j})$ newtons, acts on the particle.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

(a) Find the acceleration of the particle.

(2)

(b) At time t seconds, the velocity of the particle is $\mathbf{v}$ m s^{-1}. When $t = 0$, the velocity of the particle is $(2.2\mathbf{i} + \mathbf{j})$ m s^{-1} and the particle is at the origin.

(i) Find the distance between the particle and the origin when $t = 5$.

(4)

(ii) Express $\mathbf{v}$ in terms of t.

(2)

(iii) Find t when the particle is travelling north-east.

(3)

(Total 11 marks)

Q17.

A particle has mass 200 kg and moves on a smooth horizontal plane. A single horizontal force, $\left(400\cos\left(\frac{\pi}{2}t\right)\mathbf{i} + 600t^2\,\mathbf{j}\right)$ newtons, acts on the particle at time t seconds.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

(a) Find the acceleration of the particle at time t.

(2)

(b) When $t = 4$, the velocity of the particle is $(-3\mathbf{i} + 56\mathbf{j})$ m s^{-1}.

Find the velocity of the particle at time t.

(5)

(c) Find t when the particle is moving due west.

(3)

(d) Find the speed of the particle when it is moving due west.

(2)

(Total 12 marks)

Q18.

A football is kicked from ground level with an initial velocity of 30 m s^{-1} at an angle of 35° above the horizontal.

(a) Find the maximum height of the ball above ground level.

(4)

(b) Show that when the speed of the ball is 28 m s^{-1}, the magnitude of the vertical component of its velocity is 13.4 m s^{-1}, correct to three significant figures.

(4)

(c) Find the angle between the velocity of the ball and the horizontal when the speed of the ball is 28 m s^{-1}.

(2)

(Total 10 marks)

Q19.

Two forces, **P** = (6**i** − 3**j**) newtons and **Q** = (3**i** + 15**j**) newtons, act on a particle. The unit vectors **i** and **j** are perpendicular.

(a) Find the resultant of **P** and **Q**.

(2)

(b) Calculate the magnitude of the resultant of **P** and **Q**.

(2)

(c) When these two forces act on the particle, it has an acceleration of (1.5**i** + 2**j**) m s^{-2}. Find the mass of the particle.

(2)

(d) The particle was initially at rest at the origin.

(i) Find an expression for the position vector of the particle when the forces have been applied to the particle for t seconds.

(2)

(ii) Find the distance of the particle from the origin when the forces have been applied to the particle for 2 seconds.

(2)

(Total 10 marks)

Q20.

A particle moves on a horizontal plane, in which the unit vectors **i** and **j** are directed east and north respectively.

At time t seconds, the position vector of the particle is **r** metres, where

$$\mathbf{r} = \left(2e^{\frac{1}{2}t} - 8t + 5\right)\mathbf{i} + (t^2 - 6t)\mathbf{j}$$

(a) Find an expression for the velocity of the particle at time t.

(3)

(b) (i) Find the speed of the particle when $t = 3$.

(2)

(ii) State the direction in which the particle is travelling when $t = 3$.

(1)

(c) Find the acceleration of the particle when $t = 3$.

(3)

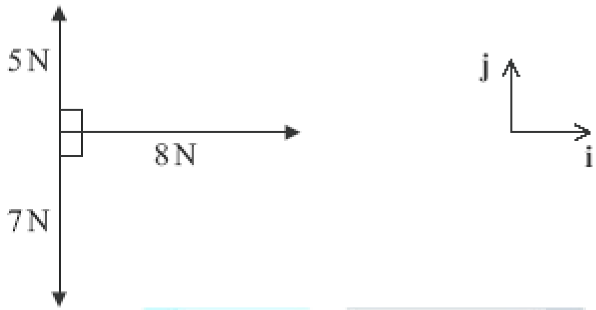
(d) The mass of the particle is 7 kg.

Find the magnitude of the resultant force on the particle when $t = 3$.

(3)
(Total 12 marks)

Q21.

The diagram shows three forces and the perpendicular unit vectors **i** and **j**, which all lie in the same plane.



(a) Express the resultant of the three forces in terms of **i** and **j**.

(2)

(b) Find the magnitude of the resultant force.

(2)

(c) Draw a diagram to show the direction of the resultant force, and find the angle that it makes with the unit vector **i**.

(3)
(Total 7 marks)

Q22.

The unit vectors **i** and **j** are directed east and north respectively. A helicopter moves horizontally with a constant acceleration of $(-0.4\mathbf{i} + 0.5\mathbf{j})$ m s^{-2}. At time $t = 0$, the helicopter is at the origin and has velocity 20**i** m s^{-1}.

(a) Write down an expression for the velocity of the helicopter at time t seconds.

(2)

(b) Find the time when the helicopter is travelling due north.

(3)

(c) Find an expression for the position vector of the helicopter at time t seconds.

(2)

(d) When $t = 100$:

(i) show that the helicopter is due north of the origin;

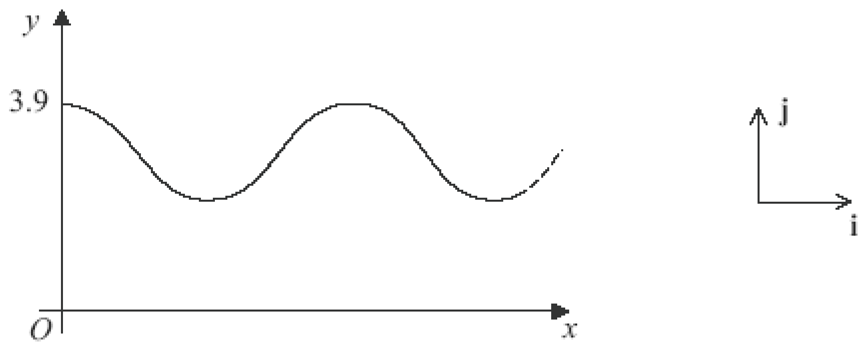
(3)

(ii) find the speed of the helicopter.

(3)
(Total 13 marks)

Q23.

Jane is on a ride in a theme park. Part of the curved path of the ride is shown in the diagram.



Jane's position vector, **r** metres, at time t seconds is given by

$$\mathbf{r} = 1.2t\,\mathbf{i} + (3 + 0.9\cos t)\mathbf{j}$$

where the perpendicular unit vectors **i** and **j** are directed horizontally and vertically upwards respectively.

(a) Find an expression for Jane's velocity at time t.

(2)

(b) (i) Find an expression for Jane's speed at time t.

(2)

(ii) Find Jane's maximum speed during the ride.

(2)
(Total 6 marks)

Q24.

A particle P moves with acceleration $(-3\mathbf{i} + 12\mathbf{j})$ m s^{-2}. Initially the velocity of P is $4\mathbf{i}$ m s^{-1}.

(a) Find the velocity of P at time t seconds.

(2)

(b) Find the speed of P when $t = 0.5$.

(3)
(Total 5 marks)

Q25.

A particle moves in a horizontal plane, in which the unit vectors **i** and **j** are directed east and north respectively. At time t seconds, its position vector, **r** metres, is given by

$$\mathbf{r} = (2t^3 - t^2 + 6)\mathbf{i} + (8 - 4t^3 + t)\mathbf{j}$$

(a) Find an expression for the velocity of the particle at time t.

(3)

(b) (i) Find the velocity of the particle when $t = \frac{1}{3}$.

(2)

(ii) State the direction in which the particle is travelling at this time.

(1)

(c) Find the acceleration of the particle when $t = 4$.

(3)

(d) The mass of the particle is 6 kg. Find the magnitude of the resultant force on the particle when $t = 4$.

(3)

(Total 12 marks)

Q26.

The unit vectors **i** and **j** are directed east and north respectively. A yacht moves with a constant acceleration. At time t seconds the position vector of the yacht is **r** metres. When $t = 0$ the velocity of the yacht is $(2\mathbf{i} - \mathbf{j})$ m s^{-1}, and when $t = 10$ the velocity of the yacht is $(-\mathbf{i} + \mathbf{j})$ m s^{-1}.

(a) Find the acceleration of the yacht.

(3)

(b) When $t = 0$ the yacht is 20 metres due east of the origin. Find an expression for **r** in terms of t.

(3)

(c) (i) Show that when $t = 20$ the yacht is due north of the origin.

(2)

(ii) Find the speed of the yacht when $t = 20$.

(4)

(Total 12 marks)

Q27.

A particle moves on a smooth horizontal surface with acceleration $(3\mathbf{i} - 5\mathbf{j})$ m s^{-2}. Initially the velocity of the particle is $4\mathbf{j}$ m s^{-1}.

(a) Find an expression for the velocity of the particle at time t seconds.

(2)

(b) Find the time when the particle is travelling in the **i** direction.

(2)

(c) Show that when $t = 4$ the speed of the particle is 20 m s^{-1}.

(4)

(Total 8 marks)

Q28.

A particle P, of mass 0.25 kg, moves in a horizontal plane so that at time t seconds its velocity $\mathbf{v}$ m s^{-1} is given by

$$\mathbf{v} = 8 \sin t\, \mathbf{i} + 4 \cos t\, \mathbf{j}$$

(a) Find the force, **F** newtons, acting on P at time t.

(3)

(b) (i) Show that the magnitude of **F** at time t is

$$\sqrt{(3\cos^2 t + 1)}$$

(3)

(ii) Hence state the range of values of the magnitude of **F**.

(2)

(Total 8 marks)

Q29.

A particle P, of mass 0.5 kg, moves under the action of a force **F** newtons, where $\mathbf{F} = 2t\mathbf{i} + 1.5\mathbf{j}$ at time t seconds.

(a) Find the acceleration of P at time t.

(1)

(b) The particle P starts its motion from rest.

Find the velocity of P at time t.

(2)

(c) The particle P starts its motion at the point O.

(i) Find the position vector of P relative to O at time t.

(3)

(ii) Find the distance of P from O when it is moving parallel to the vector $\mathbf{i} + \mathbf{j}$.

(5)

(Total 11 marks)

Q30.

Four forces, $\begin{bmatrix} 6 \\ 0 \end{bmatrix}$ newtons, $\begin{bmatrix} 0 \\ 4 \end{bmatrix}$ newtons, $\begin{bmatrix} -3 \\ -4.5 \end{bmatrix}$ newtons and $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ newtons, act on a particle.

(a) Express the resultant, **F** newtons, of these four forces as a column vector.

(2)

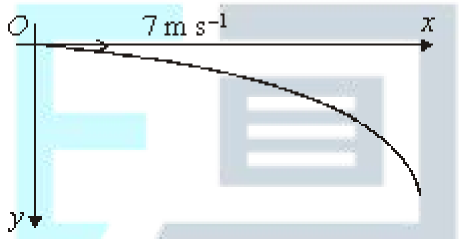
(b) Find the magnitude of **F**.

(2)

(Total 4 marks)

Q31.

Elaine kicks a ball off a cliff top. She kicks the ball from a point O and it subsequently moves in a vertical plane with respect to axes which are horizontal and vertically downwards, as shown in the diagram.



The initial velocity of the ball is horizontal and of magnitude 7 m s^{-1}.

(a) Find the coordinates of the ball t seconds after it has been kicked.

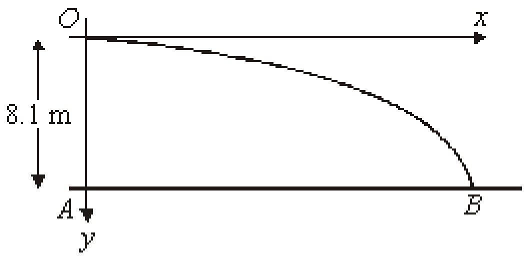
(3)

(b) Show that the equation of the path of the ball is

$$y = \frac{x^2}{10}.$$

(2)

The ball subsequently lands on a horizontal beach. The point A on the beach is vertically below O and the ball lands at the point B on the beach, as shown in the diagram.



(c) The distance OA is 8.1 metres.

Find the distance AB.

(2)

(d) Find the speed of the ball as it reaches B.

(5)

(Total 12 marks)

Q32.

A particle P moves so that at time t seconds its position vector, **r** metres, is given by

$$\mathbf{r} = 6t\mathbf{i} + t^2\mathbf{j}.$$

(a) Find the velocity of P at time t.

(2)

(b) Find an expression for the speed of P at time t.

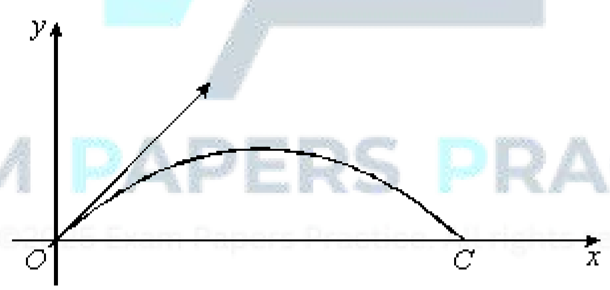
(2)

(c) Find the value of t when P moves with speed $6\sqrt{2}$ m s^{-1}.

(2)

(Total 6 marks)

Q33.



Oliver throws a ball from point O, with velocity $\begin{bmatrix} 7 \\ 7 \end{bmatrix}$ m s^{-1}. The ball is subsequently caught by

Chris at a point C which is on the same horizontal level as O, as shown in the diagram. During the motion from O to C the ball moves in a vertical plane.

(a) Taking O as the origin, find the position vector of the ball t seconds after being thrown.

(3)

(b) Show that the equation of the path of the ball is

$$y = x - 0.1x^2,$$

where (x, y) are the coordinates at time t of the ball relative to the origin O.

(4)

(c) A horizontal wire, which is part of a fence, cuts the plane of motion of the ball at the point (4, 2).

Determine whether the ball passes above or below the wire.

(3)

(d) Find the distance OC.

(4)

(e) Find the greatest speed of the ball during its motion from O to C.

(2)

(Total 16 marks)

Q34.

A particle moves so that, at time t seconds, its position vector, **r** metres, is given by

$$\mathbf{r} = 4e^{-t}\mathbf{i} + (6t + 3e^{-t})\mathbf{j}$$

The unit vectors **i** and **j** are perpendicular.

(a) Show that the velocity of the particle is $(-4\mathbf{i} + 3\mathbf{j})$ m s^{-1} when $t = 0$.

(3)

(b) Find an expression for the acceleration of the particle at time t.

(2)

(c) Find the magnitude of the acceleration of the particle when $t = 0$.

(2)

(d) Describe what happens to the velocity of the particle as t becomes large.

(2)

(Total 9 marks)

Q35.

A particle is initially at rest at the point A with position vector $(9\mathbf{i} + 10\mathbf{j})$ metres, with respect to an origin O. It moves with constant acceleration, and 10 seconds later is at the point B with position vector $(19\mathbf{i} - 25\mathbf{j})$ metres. The unit vectors **i** and **j** are perpendicular.

(a) Show that the acceleration of the particle is $(0.2\mathbf{i} - 0.7\mathbf{j})$ m s^{-2}.

(4)

(b) Find the speed of the particle at B.

(4)

(c) Verify that the particle passes through the point with position vector $(15.4\mathbf{i} - 12.4\mathbf{j})$ m.

(4)

(d) Draw a diagram to show the path of the particle as it moves from A to B. You should include the unit vectors **i** and **j** on your diagram.

(2)

(Total 14 marks)

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct vector	AO1.1b	B1	$\begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
(b)	Obtains one other edge as vector	AO1.1a	M1	$\overrightarrow{BC} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$
	Obtains $\overrightarrow{DC}$ correctly Or Obtains correctly both $\overrightarrow{BC}$ and $\overrightarrow{AD}$ Or Obtains correctly both $\overrightarrow{CB}$ and $\overrightarrow{DA}$	AO1.1b	A1	$\overrightarrow{AD} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$ $\overrightarrow{DC} = \begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
	Obtains length of one edge (or its square)	AO1.1a	M1	$AB = \sqrt{(-4)^2 + (-3)^2 + 6^2}$ $= \sqrt{61}$
	Obtains two correct lengths of different edges	AO1.1b	A1	$AD = \sqrt{1^2 + 5^2 + (-1)^2}$ $= 3\sqrt{3}$
	Completes rigorous argument to show *ABCD* is a parallelogram and not a rhombus	AO2.1	R1	$\overrightarrow{AB} = \overrightarrow{DC}$ *ABCD* must be a parallelogram $AB \neq AD$ *ABCD* is not a rhombus
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Circles correct answer	AO1.1b	B1	29
(b)	Circles correct answer	AO2.2a	B1	$90° < \theta < 135°$
				Total 2 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Sums the forces given correctly	AO1.1b	B1	$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ $= 33\mathbf{i} - 11\mathbf{j}$
	Uses Pythagoras to find the magnitude of the vector and obtains correct magnitude (given to 3 sig figs)	AO1.1b	B1	$\lvert\mathbf{F}\rvert = \sqrt{33^2 + (-11)^2}$ $= 34.8\text{ N (3 sf)}$
(ii)	Uses trig expression with appropriate values	AO1.1a	M1	$\tan\theta = \frac{11}{33}$
	Obtains correct angle (given to nearest 0.1°)	AO1.1b	A1	$\theta = \tan^{-1}\left(\frac{1}{3}\right)$ = 18.4° (3 sf) **OR** $\sin\theta = \frac{11}{\sqrt{1210}}$ $\theta = \sin^{-1}\left(\frac{11}{\sqrt{1210}}\right)$ = 18.4° (3 sf) **OR** $\cos\theta = \frac{33}{\sqrt{1210}}$ $\theta = \cos^{-1}\left(\frac{33}{\sqrt{1210}}\right)$ = 18.4° (3 sf)
(b)	States negative of 'their' part **(a)(i)**	AO2.2a	B1F	$\mathbf{F}_4 = -33\mathbf{i} + 11\mathbf{j}$
				Total 5 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$

	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their' derived values for U and V	AO3.4	M1	$v_y^2 = 8.2625^2 + 2 \times (-9.81) \times 3$
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067\ldots$
	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3\text{ m s}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part **(a)**	AO3.4	A1F	16 m s $^{-1}$
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary… air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q5.

(a) $(F =) 9\mathbf{i} - 3\mathbf{j} + 5\mathbf{i} + 8\mathbf{j} - 7\mathbf{i} + 3\mathbf{j} = 7\mathbf{i} + 8\mathbf{j}$

M1: Adding the three forces with one component correct.

A1: Correct answer.

M1A1

2

(b) $(F =) \sqrt{7^2 + 8^2} = \sqrt{113} = 10.6$ N

M1: Finding magnitude with a + sign.

A1F: Correct magnitude. Accept AWRT 10.63 and $\sqrt{113}$

Follow through incorrect answers to part (a).

M1A1F

2

(c) $(a =) \dfrac{\sqrt{113}}{5} = 2.13 \text{ m s}^{-2}$

M1: Dividing their force from part (a) or magnitude by 5.

A1F: Correct acceleration.

Accept 2.12 (from truncation or 10.6 / 5) or $\dfrac{\sqrt{113}}{5}$ *or AWRT 2.13.*

Follow through incorrect answers to parts (a) and (b).

Seeing just $\mathbf{a} = 1.4\mathbf{i} + 1.6\mathbf{j}$ *scores M1 A0*

M1A1F

2

(d) $\cos\alpha = \dfrac{7}{\sqrt{113}}$ or $\dfrac{7}{10.6}$

M1: Trig equation to find the angle with: cos with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\sin\alpha = \dfrac{8}{\sqrt{113}}$ or $\dfrac{8}{10.6}$

sin with 7 or 8 in the numerator and $\sqrt{113}$ *in denominator*

OR

$\tan\alpha = \dfrac{8}{7}$

tan with 7 and 8 in any position

A1F: Correct equation.

M1A1F

$(\alpha =) 48.8°$

A1F: Correct angle. Accept 49° or AWRT 49°

Follow through incorrect answers to parts (a) and (b).

A1F

3

[9]

Q6.

(a) $v = \int a \, \mathrm{d}t$

$= (20t^2 + t^3)\mathbf{i} - 5e^{-4t}\,\mathbf{j} + \mathbf{c}$

M1 for either term correct.
Condone no '+ ***c****'.*

M1A1

When $t = 1$,
$6\mathbf{i} - 5e^{-4}\mathbf{j} = 21\mathbf{i} - 5e^{-4}\mathbf{j} + \mathbf{c}$

Finding '+ ***c****'; not using* $\mathbf{c} = 6\mathbf{i} - 5e^{-4}\,\mathbf{j}$*.*

M1

$\mathbf{c} = -15\mathbf{i}$

A1

$\mathbf{v} = (20t^2 + t^3 - 15)\mathbf{i}\ 5e^{-4t}\,\mathbf{j}$

A1

5

(b) When $t = 0$, $\mathbf{v} = -15\mathbf{i} - 5\mathbf{j}$

M1

Speed is $\sqrt{15^2 + 5^2}$

M1

$= 15.8\text{ m s}^{-1}$

Accept $5\sqrt{10}$

A1

3

[8]

Q7.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$\mathbf{a} = (6t - 1.2\,t^2)\,\mathbf{i} + 2\,e^{-2t}\,\mathbf{j}$

A1

2

(b) $v = \int \mathbf{a}\,dt$

$= (3\,t^2 - 0.4\,t^3)\,\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$

when $t = 0$, $\mathbf{r} = 7\,\mathbf{i} - 4\,\mathbf{j}$

condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\,\mathbf{i} - 3\mathbf{j}$

$\mathbf{v} = (7 + 3\,t^2 - 0.4\,t^3)\,\mathbf{i} - (\,3 + e^{-2t})\,\mathbf{j}$

*ft from ke^{-2t} in (b); just adding 7**i** – 4**j**, m0 accept unsimplified. CAO*

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\,\mathbf{i} - 3.135\,\mathbf{j}$

ft from (b)

M1A1

speed = $\sqrt{9.6^2 + 3.135^2}$

m1

$= 10.1\ \mathrm{ms}^{-1}$

ft from (b)

A1

4

[10]

Q8.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

*M1: Using constant acceleration equation to get **r**.*

*A1: Correct expression for **r**. Allow equivalent column vector answer.*

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

*M1: Putting their **j** component equal to zero to form a quadratic equation.*

A1: Correct equation.

M1A1

t = 30 seconds

A1: For 30 seconds. No need to see t = 0.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.
A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41$ ms^{-1}

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing t = 20 and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and v = 1.41
award 4 out of 6
or
then seeing t = 20 and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q9.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\mathbf{r}$.

A1: Correct expression for $\mathbf{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\mathbf{j}$ component equal to zero to form a quadratic equation.
A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.
A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41\ \text{ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases

If the equation in t in line 2 is not seen:

then seeing t = 20 and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and v = 1.41
award 4 out of 6
or
then seeing t = 20 and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q10.

(a) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j})t$

M1: Use of constant acceleration equation to find $\mathbf{v}$ *with* $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$

A1: Correct $\mathbf{v}$*.*
(Could be done as a column vector.)

M1A1

2

(b) (i) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j}) \times 22.5$

M1: Substitution of 22.5 into their expression for the elocity, even if no marks awarded in part (a).

M1

$= -5\mathbf{i} + 5\mathbf{j}$

A1: Correct velocity CAO
(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315°.
Must follow from correct answer to (b)(i).

B1

1

(c) $(\mathbf{v} =)(4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1: Grouping $\mathbf{i}$ *and* $\mathbf{j}$ *components at some point in the solution.*
(Could be done using column vectors.)
Allow $5 = (4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

B1

$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$

M1: Seeing both components of their velocity squared and added

A1: Correct equation. (Condone including $\mathbf{i}$ *and* $\mathbf{j}$*.)*

For example: $5 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A0
$5^2 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A1

M1A1

$0.2t^2 - 3t - 8.75 = 0$

A1: Any correct simplified quadratic equation, with exactly three terms.

A1

$t^2 - 15t - 43.75 = 0$

$t = 17.5$ or $t = -2.5$

dM1: Solving the quadratic equation.

(Allow one substitution error in correctly quoted formula) Candidates with an incorrect quadratic equation must show method to get dM1.

dM1

$t = 17.5$

A1: Correct positive solution stated.

A1

6

[11]

Q11.

(a) $\mathbf{r} = \int v \, dt = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$

*M1: either **i** or **j** term correct.*
*Condone no **c***

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
$\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$

*Any attempt at **c***

M1

$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$

A1

4

(b) $\mathbf{a} = \dfrac{dv}{dt}$

$\mathbf{a} = 6t\,\mathbf{i} - 8\,\mathbf{j}$

M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$

Or: using $\mathbf{F} = m\mathbf{a}$

M1

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$= 12t\,\mathbf{i} - 16\,\mathbf{j}$

When $t = 1$, $\mathbf{F} = 12\,\mathbf{i} - 16\,\mathbf{j}$

A1

$\therefore$ Magnitude of force is $(144t^2 + 256)^{\frac{1}{2}}$ when $t = 1$

Magnitude of force is $(144 + 256)^{\frac{1}{2}}$

M1

= 20 N

= 20 N

A1

4

[10]

Q12.

(a) $\mathbf{v} = (0.5\mathbf{i} + 0.375\mathbf{j}) \times 20\ (=10\mathbf{i} + 7.5\mathbf{j})$

M1: Calculating velocity with $\boldsymbol{u} = 0\boldsymbol{i} + 0\boldsymbol{j}$ *and* $t = 20$.

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5\ \mathrm{ms^{-1}}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $$\tan\theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan\theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

θ = 053°

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$$\cos\theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos\theta = \frac{10}{12.5} \right)$$

(M1A1F)

θ = 053°

Note: For 37° award M1A0A0
But for 90 – 37 = 53° award M1A1A1.
For 127°, award M1A1A0

(A1)

OR

$$\sin\theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin\theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

θ = 053°

Condone finding angle from acceleration or position vector.

(A1)

3

(c) ($\mathbf{r}$ =) $\frac{1}{2}$ (0.5$\mathbf{i}$ + 0.375$\mathbf{j}$) t^2 (= 0.25 t^2 $\mathbf{i}$ + 0.1875t^2 $\mathbf{j}$)

M1: Finding an expression for position vector in terms of t.

A1: Correct position vector.

M1A1

$500^2 = (0.25\,t^2)^2 + (0.1875\,t^2)^2$

dM1: Using distance to form an equation for t.

A1: Correct equation.

dM1A1

$$t = 4\sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

$a = 0.625$

M1: Finding magnitude of acceleration.
A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2}0.625t^2$$

dM1: Using distance to form an equation for t.
A1: Correct equation.

(dM1A1)

$t = 40$

A1: Correct time.

(A1)

OR

$400 = \frac{1}{2}\,0.5t^2$ *or* $300 = \frac{1}{2}\,0.375t^2$

M1: Working with one component.
A1: Correct distance (300 or 400)
A1: Correct equation.

(M1A1)
(A1)

$t^2 = 1600$

dM1: Solving for t.

(dM1)

$t = 40$

A1: Correct t.
Note: 500 ÷ 12.5 = 40 is not acceptable and scores 0

(A1)

5

[12]

Q13.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1

1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1

1

(c) Due North

B1: Correct direction

B1

1

(d) $4.5\mathbf{j} = 4.5\mathbf{i} + 2\mathbf{j} + 5\mathbf{a}$

*M1: Use of a constant acceleration equation to find **a***
A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1

3

(e) $\mathbf{F} = 250\,(-0.9\mathbf{i} + 0.5\mathbf{j}) = -225\mathbf{i} + 125\mathbf{j}$

M1: Using Newton's second law.

M1

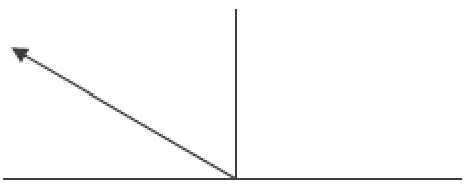
$$F = \sqrt{225^2 + 125^2} = 257\text{ N}$$

M1: Finding magnitude.
A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$\tan^{-1}\left(\frac{125}{225}\right) = 29.1°$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1°$

$180 - 29.1 = 151°$

A1: Finding required angle correctly.

A1

4

[13]

Q14.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.
A1: Correct resultant force.

M1A1

2

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or $(\mathbf{a} =) \frac{12\mathbf{i} + 8\mathbf{j}}{4}$

*M1: Use of Newton's second law with 4**a** and their answer to part (a).*

M1

$(\mathbf{a} =)\ 3\mathbf{i} + 2\mathbf{j}$ **AG**

A1: Correct acceleration from correct equation.

A1

2

(c) (i) $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + (3\mathbf{i} + 2\mathbf{j}) \times 20$

$40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + 60\mathbf{i} + 40\mathbf{j}$

B1: Seeing 60i + 40j or (3i + 2j) × 20

B1

*M1: Use of constant acceleration equation with t = 20 and **a** = 3**i** ± 2**j**.*

M1

v = (40**i** + 32**j**) – (60**i** + 40**j**)

AG

= –20**i** – 8**j**

A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.
*Note: Candidates may use **u** instead of **v** in their working.*
Example:
Starting with
***v** = 40**i** + 32**j** + (3**i** + 2**j**) × 20*
Scores B1M1A0.

A1

3

Note on Verification Method:
***v** = (–20**i** – 8**j**) + (3**i** + 2**j**) × 20 B1M1*
*= (–20 + 60)**i** + (–8 + 40)**j***
*= 40**i** + 32**j** A1*
Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) (**v** =) (–20**i** – 8**j**) + (3**i** + 2**j**)

B1: Correct velocity vector.
*Note "**v** =" does not need to be seen.*

B1

1

(iii) (**v** =) $(3t - 20)$**i** + $(2t - 8)$**j**

*M1: Velocity vector seen split into components. Condone omission of **i** and **j***
Note: This can be implied by later working, such as the second line of this solution.

M1

$$(3t-20)^2+(2t-8)^2=8^2$$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$$13t^2-152t+400=0$$

A1: Simplified quadratic equation

A1

$$t = \frac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$t = 4$ or $t = 7.69$

A1: Both correct solutions. Accept AWRT 7.7 or 7.6 or $\frac{100}{13}$.

Note: Using calculator to solve quadratic is acceptable.

A1

6

[14]

Q15.

(a) $\mathbf{v} = 9\mathbf{i} + 7\mathbf{j}$

B1: Correct initial velocity.

B1

1

(b) $\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$

M1: Writing $\mathbf{v}$ *in the form* $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= (9\mathbf{i} + 7\mathbf{j}) + (-0.01\mathbf{i} - 0.03\mathbf{j})t$

$\mathbf{a} = -0.01\mathbf{i} - 0.03\mathbf{j}$

A1: Correct acceleration.

A1

2

(c) $9 - 0.01t = -(7 - 0.03t)$

M1: Equation involving both ***i*** *and* ***j*** *components of their velocity.*
A1: Correct equation, from their acceleration.

M1A1

$$t = \frac{16}{0.04} = 400$$

A1: Correct time, from their acceleration.

A1

3

[6]

Q16.

(a) $10\mathbf{a} = 9\mathbf{i} + 12\mathbf{j}$

M1: Application of Newton's second Law with m = 10 in vector form.

M1

$\mathbf{a} = (0.9\mathbf{i} + 1.2\mathbf{j})\ \text{m s}^{-2}$

A1: Correct acceleration.
If acceleration incorrect follow their value through for the rest of this question.

A1

2

(b) (i) $\mathbf{r}(5) = (2.2\mathbf{i} + 1\mathbf{j}) \times 5 + \frac{1}{2}(0.9\mathbf{i} + 1.2\mathbf{j}) \times 5^2$

M1: Use of constant acceleration to find position vector at t = 5, with $\boldsymbol{u} \neq 0\,\boldsymbol{i} + 0\,\boldsymbol{j}$.

M1

$= 22.25\mathbf{i} + 20\mathbf{j}$

A1F: Correct position vector, for candidate's acceleration which must be a vector. Allow 22.3$\boldsymbol{i}$ + 20$\boldsymbol{j}$.

A1F

$d = \sqrt{22.25^2 + 20^2} = 29.9$ metres

dM1: Calculation of distance from position vector. Must see + sign.

dM1

A1F: Correct distance, for their acceleration. Accept 30 from 22.3$\boldsymbol{i}$ + 20$\boldsymbol{j}$.

A1F

4

(ii) $\mathbf{v} = (2.2\mathbf{i} + 1\mathbf{j}) + (0.9\mathbf{i} + 1.2\mathbf{j})t$

M1: Use of constant acceleration equation to find an expression for $\boldsymbol{v}$, with $\boldsymbol{u} \neq 0\,\boldsymbol{i} + 0\,\boldsymbol{j}$

M1

A1F: Correct $\boldsymbol{v}$ for their acceleration.

A1F

2

(iii) $\mathbf{v} = (2.2 + 0.9t)\mathbf{i} + (1 + 1.2t)\mathbf{j}$

$2.2 + 0.9t = 1 + 1.2t$

$1.2 = 0.3t$

$t = 4$

*M1: Equation involving both **i** and **j** components of their velocity. Could have incorrect signs, for example*
$2.2 + 0.9t = -(1 + 1.2t)$.

M1

A1F: Correct equation.

A1F

A1F: Correct time, for their acceleration.

A1F

3

[11]

Q17.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$400\cos\frac{\pi}{2}t\,\mathbf{i} + 600t^2\,\mathbf{j} = 200\,\mathbf{a}$

M1

$\mathbf{a} = 2\cos\frac{\pi}{2}t\,\mathbf{i} + 3t^2\,\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int a\,dt$

M1 for either $\int a\,dt$ or 1 of 2 terms correct

M1

$= \frac{4}{\pi}\sin\frac{\pi}{2}t\,\mathbf{i} + t^3\,\mathbf{j} + \mathbf{c}$

*m1 for + **c***

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,

$64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$\therefore$ $\mathbf{c} = -3\,\mathbf{i} - 8\,\mathbf{j}$

$\therefore$ $\mathbf{v} = (\frac{4}{\pi}\sin\frac{\pi}{2}t - 3)\,\mathbf{i} + (t^3 - 8)\,\mathbf{j}$

Do not accept $\frac{2}{\frac{\pi}{2}}$ *Accept 1.27 for* $\frac{4}{\pi}$

A1

5

(c) When particle is moving due west, northerly component is zero

M1

$\therefore t^3 - 8 = 0$

A1ft

$t = 2$

A1

3

(d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

B1 for change –3 to +3

B1

2

[12]

Q18.

(a) $0^2 = (30 \sin 35°)^2 + 2 \times (-9.8)s$

M1: Equation to fine the max height, with = 0

M1

A1: Correct equation

A1

M1: Solving for the height

M1

$s = \frac{(30\sin 35°)^2}{2 \times 9.8} = 15.1$ m

A1: Correct height

A1

4

(b) $28^2 = (30 \cos 35°)^2 + v_y^2$

M1: Equation to find vertical component

M1

A1: Correct equation

A1

$v_y = \sqrt{28^2 - (30 \cos 35°)^2}$

M1: Solving equation

M1

$= 13.4198 = 13.4 \text{ m s}^{-1}$ **AG**

A1: Correct speed from correct working.

A1

4

(c) $\tan\theta = \dfrac{13.4}{30 \cos 35°}$

M1: Expression to find the angle.

M1

$\theta = 28.6°$

A1: Correct angle.

A1

2

[10]

Q19.

(a) Resultant = (6**i** – 3**j**) + (3**i** + 15**j**)
= 9**i** + 12**j**

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude $= \sqrt{9^2 + 12^2}$

= 15 N

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ or $2m = 12$

$m = 6$ kg $\quad m = 6$ kg

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$\dfrac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6$ *or 6**i** + 6**j** etc without a correct previous statement gives M0A0*

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with ***u*** *= 0**i** + 0**j***

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$

$d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when $t = 2$

*(**r** = (1.5**i** + 2**j**) × 2 = 3**i** + 4**j** scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))*

M1

$= \sqrt{25} = 5$

A1: Correct distance

A1

2

[10]

Q20.

(a) $\mathbf{v} = \frac{\mathrm{d}r}{\mathrm{d}t}$

M1

$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$

***i** terms*

A1

***j** terms*

A1

3

(b) (i) When $t = 3$, $\mathbf{v} = -3.52\mathbf{i}$

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1

2

(ii) West

B1

1

(c) $\mathbf{a} = \frac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When $t = 3$, $\mathbf{a} = \frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}$ or $2.24\mathbf{i} + 2\mathbf{j}$

A1

3

(d) Using $\mathbf{F} = m\mathbf{a}$:

Accept $\mathbf{F} = 7\mathbf{a}$

M1

$\mathbf{F} = 7\left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}\right)$

$\therefore$ Magnitude of force is $7\left(\left(\frac{1}{2}e^{\frac{3}{2}}\right)^2 + 2^2\right)^{\frac{1}{2}}$

M1

$\mathbf{F} = 21.025$
$\mathbf{F} = 21.0$

Accept 21

A1

3

[12]

Q21.

(a) $\mathbf{F} = 5\mathbf{j} + 8\mathbf{i} - 7\mathbf{j} = 8\mathbf{i} - 2\mathbf{j}$

Adding the two forces. For incorrect answers, evidence of adding must be seen

M1

Correct resultant

A1

2

(b) $F = \sqrt{8^2 + 2^2} = \sqrt{68} = 8.25$ N

Finding magnitude (must see addition and not subtraction)

M1

Correct magnitude

Accept $2\sqrt{17}, \sqrt{68}$ *or AWRT 8.25 (eg 8.246)*

A1F

2

(c)

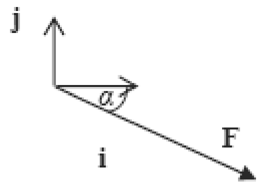


Diagram with force in the correct quadrant and with correct direction shown by an arrow.

B1

$\tan \alpha = \frac{2}{8}$

Using trig to find angle: if tan, 8 in denominator; if sin or cos, 8.25 or their answer to part (b) in denominator

M1

$\alpha = 14.0°$

Correct angle
Accept 14.1 or 14 or AWRT 14.0 (eg 14.04)
M1 and A1 not dependent on B1

A1

3

[7]

Q22.

Use of column vectors is acceptable throughout this question.

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

Use of constant acceleration equation to find expression for ***v***

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

Simplifying ***v****. (May be implied.)*
(Missing brackets may be condoned if followed by correct working.)

M1

$20 - 0.4t = 0$

*Putting **i** component equal to zero*

m1

$t = \frac{20}{0.4} = 50$ seconds

Correct time
Candidates who are able to see the correct time without supporting working gain full marks.

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

*Use of constant acceleration equation to find expression for **r***

M1

Any correct expression

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

*Substituting $t = 100$ into their expression for **r** (dependent on M1 in part (c))*

m1

$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$
$= 2500\mathbf{j}$

*Correct simplified position vector ie 2500**j***

A1

Therefore due north

Conclusion that helicopter is due north provided their position vector is of the form $k\mathbf{j}$, where $k > 0$
Note if integration is used there is no need to prove that the constant is zero.
Note marks for (d) (i) can be awarded if part (c) scores zero.

A1

3

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

*Substituting t = 100 into their expression for **v** (dependent on M1 in part (a)) or use of other constant acceleration equation and their position vector (dependent on M1 in part (c))*

m1

$= -20\mathbf{i} + 50\mathbf{j}$

Correct simplified velocity

A1

$v = \sqrt{20^2 + 50^2} = 53.9$

Correct speed (Accept $10\sqrt{29}$)

A1

Note marks for (d) (ii) can be awarded if part (a) scores zero.

3

[13]

Q23.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1

2

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1

2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$

Use of sint = 1

m1

= 1.5

A1

2

[6]

Q24.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

A1

2

(b) $t = 0.5$, $\mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

Speed = $\sqrt{(2.5^2 + 6^2}$

2 terms

M1

Speed = 6.5 ms^{1}

ft 2 terms

A1ft

3

[5]

Q25.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

substituting the value for t into their ***v***

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$ *)*

A1ft

1

(c) $\mathbf{a} = (12t - 2)\mathbf{i} - 24t\mathbf{j}$

differentiating their velocity

M1

$\mathbf{a}(4) = 46\mathbf{i} - 96\mathbf{i}$

correct acceleration at time t

A1

correct acceleration at t = 4

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$a = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q26.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

Correct ***a***

A1 3

(b) $\mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$

Use of constant acceleration equation for position

M1

$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$

*Correct **i** component*

A1

*Correct **j** component*

A1ft 3

ft incorrect acceleration

(c) (i) $\mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j}$
$= 0\mathbf{i} + 20\mathrm{j}$

*Substituting $t = 20$ into their expression for **r***

M1

so due north of origin

Correct conclusion from correct working

A1 2

(c) (ii) $\mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$

$\mathbf{v}(20) = \sqrt{4^2 + 3^2} = 5\ \mathrm{ms}^{-1}$

Finding velocity at $t = 20$

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft 4

ft incorrect acceleration

[12]

Q27.

(a) **v** = 4**j** + (3**i** – 5**j**) t

M1: Use of ***v*** *=* ***u*** *+* ***a****t and*
u *≠ 0 or Integration*
A1: Correct expression

M1A1 2

(b) **v** = 3t**i** + (4 – 5t) **j**

M1: ***j*** *component of velocity equal to zero*

M1

4 – 5t = 0
t = 0.8 s

A1: Correct t

A1 2

(c) **v** = 12**i** – 16**j**

M1: Finding velocity when t = 4

M1

v = $\sqrt{12^2 + 16^2}$ = 20 **AG**

A1: Correct velocity

A1

dM1: Finding the magnitude

dM1

A1: Correct speed from correct working

A1 4

[8]

Q28.

(a) **a** = 8 cos t **i** – 4 sin t **j**

M1 differentiation

M1A1

F = 2 cos t **i** – sin t **j**

A1 3

(b) (i) | F | = $\sqrt{(4\cos^2 t + \sin^2 t)}$

Magnitude

M1

$= \sqrt{(3\cos^2 t + 1}$

m1

CAO

A1

3

(ii) | ≤ | F | ≤ 4

B1 each value (at end of range)

B1B1

2

[8]

Q29.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4t\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1

1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

*ft **a**, both **i** and **j** present*

A1F

2

(c) (i) $\mathbf{r} = \int_0^t \left(2t^2\mathbf{i} + 3t\mathbf{j}\right)dt$

Integration attempted

M1

$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathbf{j}$

*A1 each term. Condone **i** and **j** missing.*
–1 for constants present,
–1 for both terms unsimplified

A1A1

3

(ii) $\left.\begin{array}{l}\mathbf{v} \text{ parallel to } \mathbf{i}+\mathbf{j} \\ 2t^2 = 3t\end{array}\right\}$

M1

$t = \dfrac{3}{2}$

*Solve, ft **v** with vector components*

A1F

$\mathbf{r} = \dfrac{9}{4}\mathbf{i} + \dfrac{27}{8}\mathbf{j}$

A1F

Distance $= \sqrt{\left(\left(\dfrac{9}{4}\right)^2 + \left(\dfrac{27}{8}\right)^2\right)}$

*Magnitude of **r** in any form*

M1

= 4.06 (4.056)

*Numerical answer, ft candidate's **r** and t*

A1F

5

[11]

Q30.

(a) $\mathbf{F} = \begin{pmatrix} 6 \\ -2.5 \end{pmatrix}$ N

B1 each component

B1 B1

2

(b) $|\mathbf{F}| = \sqrt{6^2 + (-2.5)^2}$

Must see +

M1

= 6.5 N

ft from vector in (a)

A1F

2

[4]

Q31.

(a) $x = 7t$

B1

$y = 0 + \frac{1}{2}gt^2$

Accept ±

M1

$y = 4.9t^2$

Accept ±

A1

3

(b) $t = \frac{x}{7}$

Attempt at substitution, or use of equation of trajectory with V = 7 & α = 0

M1

$y = 4.9\left(\frac{x}{7}\right)^2$

$y = \frac{x^2}{10}$

CAO

A1

2

(c) $8.1 = \frac{x^2}{10}$

Full method for x, accept ±

M1

$x = 9$m

AWRT 9.0 if two stages used

A1

2

(d) vert: $v^2 = 0 + 2 \times 9.8 \times 8.1$

For M1 Accept ±,
for A1 consistency of signs needed

M1A1

$u = 7$

B1

speed^2 = $(7^2 + 12.6^2)$

M1

speed = 14.4 ms^{-1}
(14.414)

A1F
5
[12]

Q32.

(a) $\mathbf{v} = 6\mathbf{i} + 2t\,\mathbf{j}$

*M1: differentiation attempted and vector quantity for **v** given*

M1A1
2

(b) sp = $\sqrt{(6^2 + 4t^2)}$ ms^{-1}

M1: sum of squares attempted giving scalar expression
A1 : all correct, accept $(2t)^2$
*f.t. **v** with 2 components*

M1A1F
2

(c) $\sqrt{(6^2 + 4t^2)} = 6\sqrt{2}$

*o.e.; scalar expression for **v** in terms of t from (b) used*

M1

$36 + 4t^2 = 36 \times 2$
$t = 3$

f.t. minor slip in (b) provided t is positive solution of quadratic eqn

A1F
2
[6]

Q33.

(a) $x = 7t$, $y = 7t - \frac{1}{2} \times 9.8t^2$ $\begin{pmatrix} 7t \\ 7t - 4.9t^2 \end{pmatrix}$

B1 horiz
M1 use of $s = ut + ½at^2$
M1A1 for vert, accept 10 or g

B1M1A1 **3**

(b) $t = \frac{x}{7}$

B1

$y = 7 \times \frac{x}{7} - \frac{1}{2} \times 9.8 \times \frac{x^2}{49}$

M1A1

$y = x - 0.1x^2$

A1 **4**

***Alternative** (full verification method)*
$x = 7t$ used B1
subs into $y = x - 0.1x^2$

M1A1

$y = 7t - 4.9t^2$ as in (a)

A1

***Alternative**: (use equation of path)*

$y = x \tan \alpha - \frac{gx^2}{2v^2 \cos^2 \alpha}$
45° used

B1

subs

M1A1

simplify

A1

(c) $x = 4$, $y = 4 - 0.1 \times 4^2$

or $t = \frac{4}{7}$ (0.571(43))

M1

$y = 2.4$

A1

above wire as 2.4 > 2

ft slip in y provided y positive

A1F

3

(d) $y = 0$

$x - 0.1x^2 = 0$

M1

$x(1 - 0.1x) = 0$

m1 A1

$x = 10$

A1F

4

***Alternative**: (use of coordinate equations)*

$y = 0$, $7t - 4.9t^2 = 0$

M1

$\Rightarrow t = \frac{10}{7}$ *(1.43)*

m1A1

$x = 7t$ $= 10$

A1F

(Accept 10.01 if t = 1.43 used)

***Alternative**: (use of symmetry)*

$v = 0$, $0 = 7 - 9.8t$

M1

$t = \frac{5}{7}$

A1

$x = 7 \times t \times 2$

m1

$x = 10$ (acc 9.996)

A1F

***Alternative**: (use of equation for Range)*

$R = \dfrac{2V^2 \sin\alpha\cos\alpha}{g}$

Correct formula used M1, 45° used m1
Correct subs A1,
result A1F, AWRT 10m, ft one slip.

(e) Speed^2 = $7^2 + 7^2$

M1

= 9.90 ms^{-1}

Accept 9.9, 7√2,
or 9.91 if t = 1.43 used in downward motion

A1

2

[16]

Q34.

(a) $\mathbf{v} = -4e^{-t}\,\mathbf{i} + (6 - 3e^{-t})\mathbf{j}$

Differentiating position vector

M1

$t = 0$ A1

Correct velocity

$\mathbf{v} = -4\mathbf{i} + 3\mathbf{j}$

ag Substituting t = 0 to obtain initial velocity

A1

3

(b) $\mathbf{a} = 4e^{-t}\,\mathbf{i} + 3e^{-t}\mathbf{j}$

Differentiating velocity

M1

Correct acceleration

A1

2

(c) $\mathbf{a} = 4\mathbf{i} + 3\mathbf{j}$

Finding acceleration when t = 0

M1

$a = \sqrt{4^2 + 3^2} = 5$

Correct magnitute

A1

2

(d) $\mathbf{v} \to 0\mathbf{i} + 6\mathbf{j}$

*For **i** component*

B1

*For **j** component*

B1

2

[9]

Q35.

(a) $19\mathbf{i} - 25\mathbf{j} = \frac{1}{2}\mathbf{a} \times 10^2 + 9\mathbf{i} + 10\mathbf{j}$

Using both position vectors to form a constant acceleration equation

M1

$50\mathbf{a} = 10\mathbf{i} - 35\mathbf{j}$

Correct equation

*Solving for **a***

A1 M1

$\mathbf{a} = 0.2\mathbf{i} - 0.7\mathbf{j}$

*Correct **a***

A1

4

(b) $\mathbf{v} = 10(0.2\mathbf{i} - 0.7\mathbf{j})$

*use of **v** = **a**t*

M1

$= 2\mathbf{i} - 7\mathbf{j}$

Correct velocity

A1

$v = \sqrt{2^2 + 7^2} = 7.28 \text{ ms}^{-1}$

Finding speed from velocity

Correct speed

m1 A1

4

(c) $15.4 = \frac{1}{2} \times 0.2 \times t^2 + 9$

Finding t from one component

M1

$t = 8$

Correct t

A1

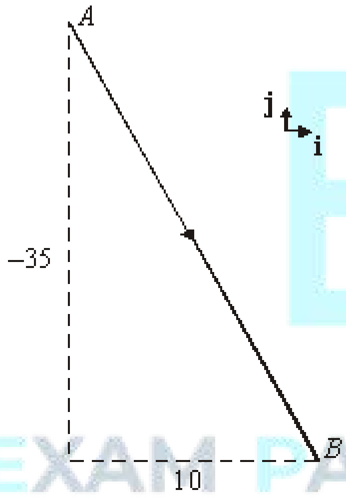
$\frac{1}{2} \times (-0.7) \times 8^2 + 10 = -12.4$

Using t = 8 with other component
Correct result

M1 A

4

(d)



Straight line
Correct direction

B1 B1

2

[14]

Examiner reports

Q1.

Question (a) was completed successfully by most students. The most common error was a sign mistake.

Well over half of the students scored at least two marks on part (b) The best students realised they had to do very little to answer this question and, provided they explained their arguments rigorously, they achieved full marks. Most students completed more calculations than necessary (there was no need to find the lengths of all sides), but failed to complete their arguments in sufficient detail. For example, they may have explained why the shape was a parallelogram, but not explained why it was not a rhombus.

Q5.

Again this question was done well by many students although some made minor errors. In part (c) a few students found the acceleration (as $1.4\mathbf{i} + 1.6\mathbf{j}$) and did not calculate the magnitude. Also, in part (d) some students found the angle as 41° rather than 49°. However many students gained full marks on this question.

Q6.

This question was also answered well by the vast majority of students, who showed that they knew the techniques involved in this question. In part (a) the majority of students integrated to find the velocity of the particle but a common error was to forget the + $\mathbf{c}$ term and thus to ignore the initial position vector. Some found the + $\mathbf{c}$ incorrectly, assuming that $\mathbf{c}$ was $6\mathbf{i} - 5e^{-4}\,\mathbf{j}$ without completing the necessary calculation.
Another common error was seen whereby students did not integrate the e^{-4t} term correctly.
In part (b) a number of students found the initial velocity, $-15\mathbf{i} - 5\mathbf{j}$, but did not find the initial speed.

Q7.

This question was also answered well. In part (a), a few candidates did not divide **F** by 50 accurately. In part (b), a number of candidates did not find their '+$\mathbf{c}$' correctly, simply replacing their '$\mathbf{c}$' with $7\mathbf{i} - 4\mathbf{j}$. Others, on evaluating their '$\mathbf{c}$', took $-e^{-2t}$ when $t = 0$ to be +1 rather than –1.

Q8.

There were some vary varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).
Part (c) was altogether more challenging and many candidates worked with position vectors rather than velocity vectors. This clearly was the wrong approach, but some expended a lot of effort solving the quadratic equations that they produced. Some candidates came up with the correct velocity, but without showing how they had obtained their answer. It is important that working is justified. Sometimes a trial and improvement

method led candidates to the correct answer.

Q9.

There were some very varied responses to this question. Part (a) was often done well and the candidates who attempted to find the position vector generally gained the marks available. However, quite a few gave the velocity instead of the position vector. Those who had a velocity vector in part (a) then generally used this in part (b) and scored no marks. Some who had a correct position vector in part (a), used the wrong component to try to answer part (b).
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Q10.

Parts (a) and (b) were generally done well, with the candidates gaining many marks. A few candidates could not state the direction in part (b)(ii) and gave answers such as north-east or south-west.Part (c) was found to be more demanding. Candidates who did not identify the **i** and **j** components of the velocity made very little progress. Some made some initial good progress, but made arithmetic or algebraic errors in their solutions. Some forgot to square the speed and tried to work from equations such as $(4 - 0.4t)^2 + (0.5 + 0.2t)^2 = 5$. Some candidates used a trial and improvement approach and realised that the velocity must be $-3\mathbf{i} + 4\mathbf{j}$, which produced a speed of 5.

Q11.

Most candidates answered this question well. In part (a), a small number of candidates forgot to add, and find, the +**c** term and thus ignored the initial position vector; a few candidates tried to use equations for constant acceleration. In part (c), some candidates made errors when finding the force by finding $m\mathbf{a}$, which was $2(6t\mathbf{i} - 8\mathbf{j})$, to be $12t\mathbf{i} - 8\mathbf{j}$.

Q12.

In part (a), many candidates produced correct answers, but some stopped when they had obtained the velocity and did not go on to find the speed as requested. Part (b) was generally done well, but a few candidates did not give their final answer to the nearest degree. Part (c) was very much more demanding. A significant number of candidates mixed scalar and vector quantities in an attempt to answer the question. Statements like $500 = \frac{1}{2}(0.5\mathbf{i} + 0.375\mathbf{j})t^2$ were quite common and there were some similar ones that also included an initial velocity. The successful candidates used a variety of approaches, which can be seen in the mark scheme. The key difference between those who were successful and those who were not was the ability to connect the scalar distance of 500 to the position vector in a meaningful way.

Q13.

Most candidates answered parts (a), (b) and (c) well, although some did get an answer of

–4.5**j** for part (b).

In part (c), the candidates who clearly stated and used a constant acceleration equation usually made a good start and completed the question, although there were a few arithmetic errors.

In part (e) many candidates did not attempt to find the magnitude of the force.

Part (f) was generally found more difficult with only a few good attempts. Several candidates did not attempt this part of the question.

Q14.

Parts (a) and (b) of this question were done very well and many candidates gained all four marks.

Part (c) proved to be more challenging. In part (c)(i), many candidates did not show how they started their working. An equation based on $\mathbf{v} = \mathbf{u} + \mathbf{a}t$ was expected, but not always seen. Some candidates obtained the printed answer, but did not gain full marks as their starting point was not clear. Having both the acceleration and the initial velocity given in the question enabled many candidates to write down a correct expression for the velocity of the particle. Part (c)(iii) was found to be very challenging. If candidates were able to write down an expression for the speed of the particle they would often go on to find the times correctly. It was this vital first step that many candidates were unable to take. A few candidates made errors solving the quadratic equation once they had obtained it correctly.

Q15.

The majority of the candidates were able to find the initial velocity as requested in part (a), but most found the rest of the question difficult. A reasonable number of the candidates were able to find the acceleration, some by calculating two velocities and using $\mathbf{v} = \mathbf{u} + \mathbf{a}t$, but others clearly could not find a strategy to find the acceleration. Part (c) was found to be more demanding. Some candidates did use the components correctly, but quite a number of candidates did not recognise the approach that was required.

Q16.

While some candidates did well with this question, others made some fundamental errors. In part (a), some candidates found the magnitude of the acceleration and then tried to use this scalar quantity in the later parts of the question, ending up with a mix of scalar and vector quantities.

In part (b)(i), many candidates found the position vector, but did not go on to find the distance as requested. There were quite a number of correct responses to part (b)(ii), but quite often this was not followed by a correct approach to the final part of the question.

Q17.

Virtually all candidates answered part (a) correctly. The common problem in part (b) was in the integration of $400\cos\frac{\pi}{2}t$. Apart from this, answers to part (b) were good, and most correctly found $t = 2$ in part (c). The majority of candidates also found correctly that the speed of the particle was 3, with only a few finishing with the velocity of the particle being

–3 or –3**i**.

Q18.

This question also provided more challenge for the candidates. In this question a number of candidates did assume that the acceleration due to gravity was positive.

In part (a), quite a number of candidates found the time of flight and then used half of this value to calculate the maximum height. There were however a number of cases where the candidates did not half their value for the time of flight and so produced solutions which did not relate to the maximum height.

Part (b) was much more demanding and relatively few correct solutions were seen. Those candidates who identified the required method were able to do well, but the majority made no attempt or produced working which could not lead to the required answer. Similarly part (c) was found to quite difficult, but some candidates did use the answer from part (b) to put together correct solutions for part (c) when they had been unable to do part (b).

Q19.

Parts (a) and (b) were generally done well by the candidates. A few candidates subtracted the vectors in part (a) instead of adding them. Also some tried to find the magnitude of each vector in part (a) and then lost several marks because they had very confused working.

In part (c) there were many attempts which involved the division of vectors. For example working such as: $m = \dfrac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6\mathbf{i} + 6\mathbf{j}$. Candidates should be advised to consider the components involved and not to attempt to divide vectors.

There were a number of errors that appeared in part (d). One was to include an initial velocity, which was equal to the resultant force on the particle. A few candidates used incorrect formulae. In the final part of the question a large number of candidates gave the vector 3**i** + 4**j** as their final answer and did not find the magnitude in order to get the distance as requested.

Q20.

Most of this question was answered well. In part (a), the only difficulty was found in differentiating $2e^{\frac{1}{2}t}$. In part (b), many found the velocity of the particle to be $(e^{\frac{3}{2}} - 8)\mathbf{i}$ [or –3.52**i**], but did not find the speed of +3.52 m s^{-1}. Others gave the speed as 3.5 m s^{-1}, which was penalised. Those candidates who were successful in part (a) usually completed parts (c) and (d) correctly.

Q21.

This question was also done well by many candidates. In part (a), there were some confused responses and occasional problems with minus signs. Part (b) was also generally done well. In part (c), a few candidates had problems. The most common of these was to omit an arrow from the diagram to show the force; the other was to use the lengths incorrectly when calculating the angle, for example $\tan^{-1}\left(\frac{8}{2}\right)$ and $\sin^{-1}\left(\frac{8}{8.25}\right)$.

Q22.

Generally the candidates made quite good attempts at this question, but some parts caused the candidates more difficulty. A few candidates made poor use of vector notation. Part (a) was answered correctly by the vast majority of the candidates. The most common error was dealing incorrectly with the initial velocity. A few went on to simplify their answers incorrectly. Part (b) was a little more demanding, as some candidates confused the components. Part (c) did cause a little more difficulty for some candidates, with a few candidates not giving any answer at all. However, there were again many correct solutions.

In part (d)(i), although there were many correct solutions, some candidates showed that the helicopter was south or north of the origin, rather than specifically north. Interestingly, some candidates who could not answer part (c) were able to write down the position vector at the specified time. A few candidates found the correct position vector but did not conclude that the helicopter was due north.

When answering part (d)(ii), many of the candidates found the correct velocity, but some did not find the speed as requested.

Q23.

While candidates realised that part (a) required differentiation, a significant number were not able to carry this out accurately. Part (b) proved more successful, but only a small minority of candidates were able to attempt part (c), the simplest solution involving maximisation of the appropriate trigonometrical function.

Q24.

Part (a) was done well, but in part (b) many stopped after finding the velocity at the given time, instead of continuing to find the speed by evaluating the magnitude of the velocity vector.

Q25.

There were very many good solutions to this question and a good number of candidates scored full marks. Some candidates did however make errors. In part (a) there were some cases where the candidates made some minor errors when differentiating one or both of the components. In part (b) a few candidates had difficulties substituting t = 1/3 correctly, but the biggest problem in this part of the question was the description of the direction. While a number did state that the direction was south, some stated that it was north, even though they had calculated the velocity correctly. Others did not use either north or south, but gave answers that implied that **j** was a vertical unit vector.

Parts (c) and (d) were generally done well, but a few candidates experienced difficulties with the differentiation and very occasionally with the substitution of t = 4 .

Q26.

Candidates found this question quite demanding. Candidates were most successful with part (a) and for some this was the only place where they gained marks. In part (b), many candidates omitted the initial position of the yacht, and a number tried to integrate the initial velocity. In part (c)(i), a poor answer for the position vector had an impact. However, some candidates who had good expressions simply showed that the i component of the position was zero and not that the yacht was due north of the origin. In part (c)(ii), quite a few candidates found the velocity correctly, but did not go on to find the speed.

Q27.

There were many good responses to this question. Generally part (a) was done very well. Part (b) was most demanding, but many candidates still did well on this part. One error was to work with a position vector instead of a velocity vector. In part (c), a few candidates gave the velocity as their final answer instead of the speed.

Q28.

Part (a) was frequently done well, with the most common error in part (a)(ii) being the introduction into the equation of motion of an additional force, usually the weight of the motorcycle and rider. Part (b) seldom produced assumptions backed by sound reasons. The majority of responses referred to information given in the question. In part (c), the most convincing solutions were based on the algebraic link between the force and the radius. There was a tendency in part (c) to assume that the magnitude of the frictional force could be altered by changing the radius of the circle, rather than the necessity to adjust the radius to compensate for the force.

Q29.

For those who were well prepared in the calculus techniques linking the quantities involved, this question proved highly successful. Part (c)(ii) proved discriminating, as expected, with only the most able candidates linking the components of the velocity vector with the given vector in an appropriate way.

Q30.

This was a popular starter question often yielding full marks, with only occasional careless errors in arithmetic in part (a) and signs in part (b).

Q31.

It was refreshing to see a wide variety of approaches to this question, which differed in many ways from previous questions on this topic. Many candidates were able to find the coordinates of the ball in part (a) and at this stage an error in the direction of the y-value was condoned. Part (b) provided interesting responses, using either the answers to part (a) with subsequent elimination of the time, or, in some cases, the full equation of the trajectory. In part (c) many candidates used the result of part (b), while others successfully found the time involved as an intermediate step. Part (d) proved to be more discriminating and there was confusion, for some, as to the quantities involved in the motion in the two directions. Many candidates reached the correct result following sound work, but some lost a mark through failing to adhere to the degree of accuracy required by the rubric for the paper. Overall it is encouraging to see candidates who are unable to complete one part of a question continue to other parts where they can be more successful.

Q32.

Most could attempt differentiation correctly to find an expression for the velocity in part (a).

However poor algebra and lack of appreciation of the form of scalar quantities let down many in part (b). Correct answers in part (b) were often then reduced to '6 + 2t' in part (c) and few completed the request correctly.

Q33.

The quality of solutions to problems set on this area of the specification has improved steadily since the first paper in 2001. It was pleasing to see a variety of methods being used and many candidates scored well on this demanding question.

Part (a) was attempted successfully either using the equations of constant acceleration or by integration. The most common error was an incorrect sign in the acceleration of the vertical component.

Part (b) proved to be the most demanding part of the question. Again, varied methods were used, ranging from a full verification of the result to the quotation of the general equation of the path of the trajectory with appropriate substitutions.

Part (c) was popular, yielding marks to most candidates.

The varied methods used in part (d) revealed clear understanding of the motion. The most popular methods used the result quoted in part (b) or the equations of motion with constant acceleration, sometimes incorporating the symmetry of the motion.

In part (e), for those candidates who realised that the greatest speed occurred either at O or at C, the request was straightforward and full marks were often obtained.

Q34.

Parts (a), (b) and (c) were generally answered very well.

In part (a), a few candidates tried to substitute t = 0 into the position vector without differentiating.

In part (c), some candidates found the acceleration, but did not go on to find the magnitude.

Part (d) was more challenging. Quite a few candidates stated that the velocity would become zero. There were some very good answers, but also quite a few that were somewhat confused.

Q35.

The candidates who chose the appropriate constant acceleration equation in part (a) had little difficulty. However, a number did treat the position vector as if it were a velocity vector. In part (b), some candidates stopped when they had found the velocity and did not go on to find the speed as requested. A few candidates had problems dealing with the negative signs. Part (c) was very challenging. Some candidates made good attempts at this, but they were very much in the minority. There were quite a few reasonable answers to part (d).